Inverse & Composite Functions

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Composing functions

Given $f(x) = \sqrt{x+4}$ and $g(x) = x^2 - 4$, find $(f \circ g)(2)$ and state the domain restriction that makes $f \circ g$ defined.

2 [Foundational] • Target: Decomposing a composite function

Write $h(x) = (3x - 5)^4$ as a composition $h(x) = f(g(x))$ for two simpler functions f and g.

3 [Foundational] • Target: Decomposing a composite function

Write $h(x) = \sqrt{x^2 + 1}$ as a composition of two functions.

4 [Foundational] • Target: Verifying inverse functions

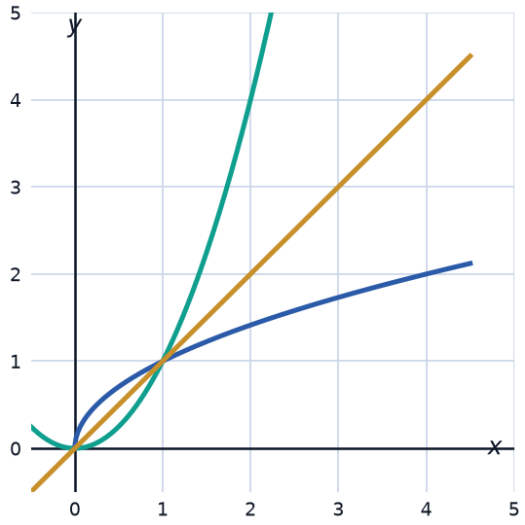
Verify algebraically that $f(x) = \frac{x-2}{3}$ and $g(x) = 3x + 2$ are inverses of each other.

5 [Foundational] • Target: Verifying inverse functions

Find the inverse of $f(x) = 2x^3 + 1$, and verify that $f(f^{-1}(x)) = x$ for one value, say $x = 17$.

6 [Foundational] • Target: Visualizing inverse functions as reflections

The graph shows $f(x) = x^2$ (restricted to $x \geq 0$) and its inverse $f^{-1}(x) = \sqrt{x}$, which are reflections of each other across the line $y = x$.



a State the domain restriction placed on $f(x) = x^2$ needed for it to have an inverse, and explain why.

b Confirm from the graph that f and f^{-1} intersect on the line $y = x$, and find that intersection point algebraically.

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7 [Foundational] • Target: Domain and range of composite and inverse functions

Given $f(x) = \frac{1}{x-3}$ and $g(x) = x + 1$, find the domain of $(f \circ g)(x)$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Domain and range of composite and inverse functions

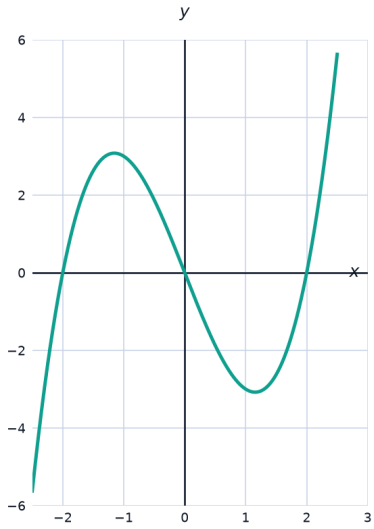
If f has domain $[1, \infty)$ and range $[0, \infty)$, state the domain and range of f^{-1}.

9 [Intermediate] • Target: Applying composite and inverse functions in context

A temperature-conversion function $C(F) = \frac{5}{9}(F - 32)$ converts Fahrenheit to Celsius. Find $C^{-1}(F)$ and interpret what it computes.

10 [Intermediate] • Target: Using the horizontal line test

Determine whether $f(x) = x^3 - 4x$ has an inverse function over its entire domain. Justify using the horizontal line test.



11 [Intermediate] • Target: Using the horizontal line test

Explain why $f(x) = 5x - 2$ passes the horizontal line test for all real x, and find its inverse.

12 [Intermediate] • Target: Composite functions with restricted domains

Given $f(x) = \frac{1}{x}$ and $g(x) = x - 4$, find $(f \circ g)(x)$ and state its domain.

13 [Intermediate] • Target: Domain and range of composite and inverse functions

Find the inverse function of $f(x) = \frac{2x-1}{x+3}$.

14 [Intermediate] • Target: Applying composite and inverse functions in context

Determine whether the function $g(x) = \frac{x^3}{x^2+4}$ is even, odd, or neither.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Using the horizontal line test** • **Cross-Lesson: 2.01 Exponential Functions & Growth Models**

A function is defined piecewise: $f(x) = x^2$ for $x < 0$, and $f(x) = 2x + 1$ for $x \geq 0$. What is the value of $f(-3) + f(2)$?

16 **[Challenging]** • **Target: Composite functions with restricted domains** • **Cross-Lesson: 2.02 Logarithmic Functions & Properties**

[Rational Functions, Limits & Continuity · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

17 **[Challenging]** • **Target: Composing functions** • **Cross-Lesson: 3.01 Trigonometric Functions & the Unit Circle**

[Exponential & Logarithmic Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

18 **[Challenging]** • **Target: Decomposing a composite function** • **Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs**

[Functions & Their Graphs · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

19 **[Challenging]** • **Target: Verifying inverse functions** • **Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions**

[Polynomial Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

20 **[Challenging]** • **Target: Visualizing inverse functions as reflections** • **Cross-Lesson: 4.03 Parametric Equations & Modeling**

[Trigonometric Functions · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 **[Foundational]** • **Target: Composing functions**

Given $f(x) = -7x$, find the slope and y-intercept of $f^{-1}(x)$ as an ordered pair (slope, intercept).

a. $(-7, 0)$

b. $\left(-\frac{1}{7}, 0\right)$

c. $\left(\frac{1}{7}, 0\right)$

d. $\left(\frac{5}{14}, 0\right)$

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22 [Foundational] • Target: Decomposing a composite function

Determine whether $f(x) = x^3 - 6x$ has an inverse function over its entire domain, using the horizontal line test.

a. Has an inverse function over its entire domain (strictly increasing)

b. Does NOT have an inverse function over its entire domain (not one-to-one)

c. Has an inverse only on the interval $x \geq 0$

d. Cannot be determined without a graph

23 [Foundational] • Target: Verifying inverse functions

What is the domain and range of $y = \tan^{-1}(x)$?

a. Domain $(-\infty, \infty)$, range $(-\infty, \infty)$

b. Domain $(-\infty, \infty)$, range $[0, \pi]$

c. Domain $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, range $(-\infty, \infty)$

d. Domain $(-\infty, \infty)$, range $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

24 [Foundational] • Target: Visualizing inverse functions as reflections

What is the domain and range of $y = \sin^{-1}(x)$? (variant 5)

a. Domain $[-1, 1]$, range $(-\infty, \infty)$

b. Domain $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, range $[-1, 1]$

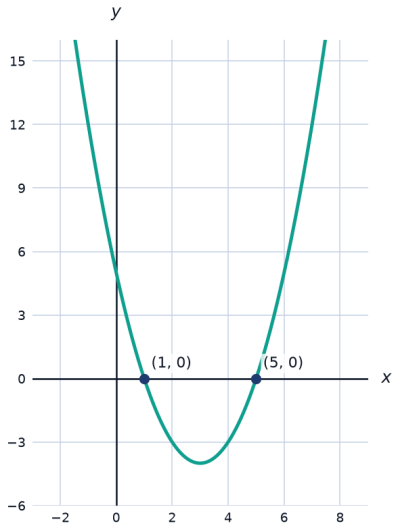
c. Domain $[-1, 1]$, range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

d. Domain $(-\infty, \infty)$, range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

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25 [Foundational] • Target: Domain and range of composite and inverse functions

The graph of the quadratic function $f(x) = x^2 - 6x + 5$ is shown in the standard (x, y) coordinate plane below. A horizontal line $y = k$ intersects the graph of $f(x)$ at two points that are 8 units apart. What is the value of k?



a. 12

b. 8

c. 4

d. -3

26 [Foundational] • Target: Applying composite and inverse functions in context

Consider the function $f(x) = 2x + 5$. Which of the following represents the inverse function $f^{-1}(x)$?

a. $f^{-1}(x) = \frac{x-5}{2}$

b. $f^{-1}(x) = \frac{x+5}{2}$

c. $f^{-1}(x) = \frac{2}{x} - 5$

d. $f^{-1}(x) = 2x - 5$

27 [Foundational] • Target: Using the horizontal line test

Find the inverse function: if $y = \sin x$, then $x =$

a. $\sin^{-1} y$ for $y \in [0, 1]$ only

b. $\cos^{-1} y$

c. π/y

d. $\arcsin y$ for $y \in [-1, 1]$

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28 [Foundational] • Target: Composite functions with restricted domains

A ramp rises 3 m over a horizontal distance of 20 m. Find its angle of inclination, to the nearest tenth of a degree.

a. 8.5°

b. 81.5°

c. 13.5°

d. 17.1°

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Composing functions

Find the inverse of $f(x) = \frac{3x+2}{x-1}$ and state its domain.

a. $f^{-1}(x) = \frac{x+2}{x-3}$; domain: all reals except $x = 3$

b. $f^{-1}(x) = \frac{3x-2}{x+1}$

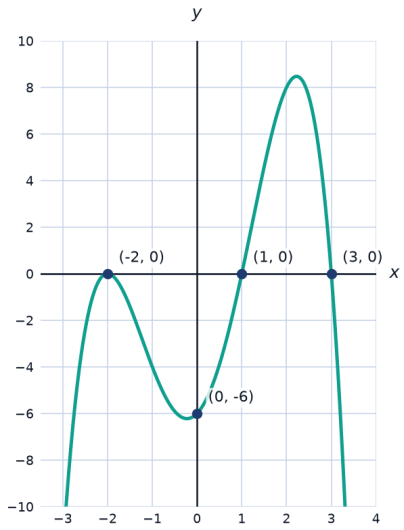
c. $f^{-1}(x) = \frac{x+2}{x+3}$

d. $f^{-1}(x) = \frac{x-2}{x+3}$

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30 **[Intermediate]** • **Target: Decomposing a composite function**

The graph of a polynomial function $f(x)$ of degree 4 is shown in the figure below. The graph is tangent to the x-axis at $(-2, 0)$, intersects the x-axis at $(1, 0)$ and $(3, 0)$, and crosses the y-axis at $(0, -6)$. A new polynomial $Q(x)$ is defined by $Q(x) = f(x) + \frac{1}{2}x^3$. When $Q(x)$ is divided by $x^2 - 4$, the remainder is the linear polynomial $R(x)$. What is the value of $R(5)$?



a. 24

b. 18

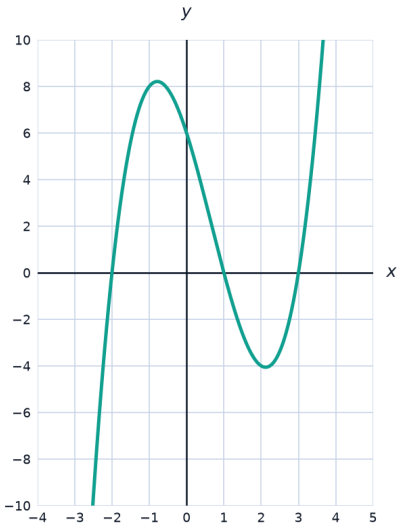
c. 17

d. 14

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31 **[Intermediate]** • **Target: Verifying inverse functions**

The graph of a polynomial function $f(x)$ is shown below. Which of the following could be the equation for $f(x)$?



a. $f(x) = (x + 2)(x - 1)(x - 3)$

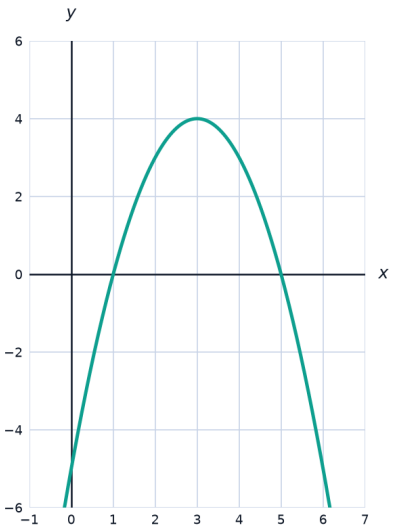
b. $f(x) = (x - 2)(x + 1)(x + 3)$

c. $f(x) = -(x + 2)(x - 1)(x - 3)$

d. $f(x) = (x + 2)(x - 1)^2(x - 3)$

32 **[Intermediate]** • **Target: Visualizing inverse functions as reflections**

The graph of the function $f(x) = -(x - 3)^2 + 4$ is shown in the xy-plane below. What is the maximum value of $f(x)$?



a. -5

b. 3

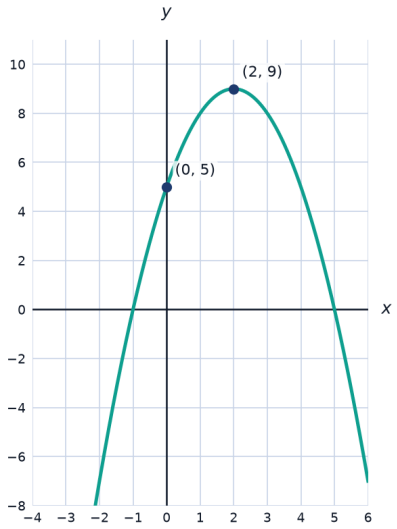
c. 4

d. 7

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33 **[Intermediate]** • **Target: Domain and range of composite and inverse functions**

The graph of the quadratic function $f(x) = a(x - h)^2 + k$ is shown in the xy-plane with vertex $(2, 9)$ and y-intercept $(0, 5)$. A linear function is defined by $g(x) = mx + 1$. If the graph of g passes through the vertex of the graph of f, what is the other x-coordinate where the graphs of f and g intersect?



a. -4

b. -2

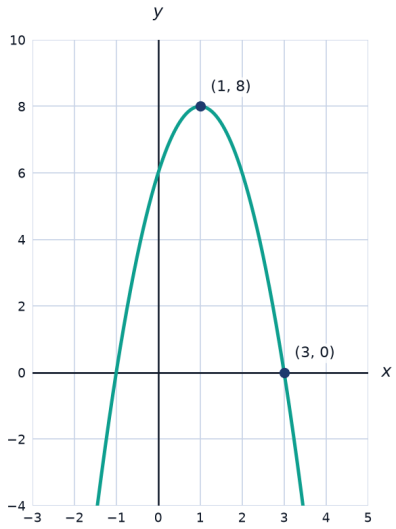
c. -1

d. 1

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34 **[Intermediate]** • **Target: Applying composite and inverse functions in context**

The graph of the quadratic function $f(x)$ is shown in the figure, with its vertex at $(1, 8)$ and an x-intercept at $(3, 0)$. A second function g is defined by $g(x) = f(x - 2) + k$, where k is a constant. If the graph of g passes through the origin $(0, 0)$, what is the value of k?



a. -10

b. -6

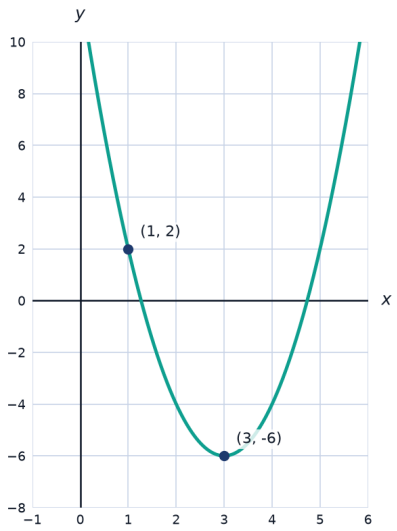
c. 6

d. 10

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35 [Intermediate] • Target: Using the horizontal line test

The quadratic function $f(x) = ax^2 + bx + c$ is graphed in the standard (x, y) coordinate plane below. The vertex of the parabola is at $(3, -6)$, and the graph passes through the point $(1, 2)$. What is the value of b?



a. -12

b. -6

c. 6

d. 12

36 [Intermediate] • Target: Composite functions with restricted domains

Consider the function $f(x) = 4\cos^2(x) + b\sin(x) + c$ where b and c are real constants with $b > 0$. If the maximum value of $f(x)$ is 10 and the minimum value of $f(x)$ is 1, what is the value of c?

a. 5

b. $\frac{11}{2}$

c. 9

d. 4

37 [Intermediate] • Target: Composing functions

The function $f(x) = a\cos(bx) + d$, where $a, b,$ and d are positive constants, has a period of π, a maximum value of 7, and satisfies $f(\frac{\pi}{6}) = 4$. What is the value of $f(\frac{\pi}{3})$?

a. -2

b. 1

c. 4

d. -5

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Decomposing a composite function • Cross-Lesson: 3.02 Polar Coordinates & Polar Graphs

Consider the rational function $f(x) = \frac{x^2 - 2x + c}{x^2 + 2x + c}$ where c is a real constant. If the domain of f is the set of all real numbers and the maximum value of $f(x)$ is 4, what is the value of c?

a. $\frac{25}{9}$

b. $\frac{9}{4}$

c. 3

d. $\frac{16}{5}$

39 [Challenging] • Target: Verifying inverse functions • Cross-Lesson: 4.02 Piecewise-Defined & Implicit Functions

A function f is defined for all real numbers x such that $x \neq 0$ and $x \neq 1$. If $f(x) + 2f\left(\frac{1}{1-x}\right) = 3x$ for all permissible values of x, what is the value of $f(3)$?

a. $\frac{20}{9}$

b. $\frac{7}{3}$

c. $\frac{16}{9}$

d. 3

40 [Challenging] • Target: Visualizing inverse functions as reflections • Cross-Lesson: 4.03 Parametric Equations & Modeling

Which of the following statements describes the end behavior of the polynomial function $f(x) = -5x^4 + 2x^3 - 7x + 3$?

a. As $x \to -\infty$, $f(x) \to -\infty$, and as $x \to \infty$, $f(x) \to -\infty$

b. As $x \to -\infty$, $f(x) \to \infty$, and as $x \to \infty$, $f(x) \to -\infty$

c. As $x \to -\infty$, $f(x) \to -\infty$, and as $x \to \infty$, $f(x) \to \infty$

d. As $x \to -\infty$, $f(x) \to \infty$, and as $x \to \infty$, $f(x) \to \infty$

41 [Challenging] • Target: Domain and range of composite and inverse functions • Cross-Lesson: 1.01 Polynomial End Behavior & Zeros

A parabola has its vertex at the origin $(0, 0)$ and passes through the point $(2, -4)$. If the directrix of the parabola is a horizontal line, what is the equation of the directrix?

a. y = 4

b. y = -4

c. x = 4

d. x = -4

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42 [Challenging] • Target: Applying composite and inverse functions in context • Cross-Lesson: 1.02 Rational Functions & Asymptotes

A factory produces two types of components, A and B, using two machines, M1 and M2. The time (in minutes) required to produce one unit of each component on each machine is given in the table below: | Component | Machine M1 | Machine M2 | |---|---|---| | A | 2 | 3 | | B | 4 | 1 | The factory has 480 minutes of M1 time and 240 minutes of M2 time available per day. Let x be the number of units of component A produced per day and y be the number of units of component B produced per day. Suppose the profit from selling one unit of component A is 5 and the profit from selling one unit of component B is 8. The factory wants to maximize its total daily profit. Formulate this problem as a linear programming problem, defining the objective function and the constraints.

a. Maximize $P = 5x + 8y$ subject to $2x + 4y \le 480$, $3x + y \le 240$, $x \ge 0$, $y \ge 0$.

b. Maximize $P = 8x + 5y$ subject to $2x + 3y \le 480$, $4x + y \le 240$, $x \ge 0$, $y \ge 0$.

c. Maximize $P = 5x + 8y$ subject to $4x + 2y \le 480$, $y + 3x \le 240$, $x \ge 0$, $y \ge 0$.

d. Maximize $P = 5x + 8y$ subject to $2x + 4y \le 240$, $3x + y \le 480$, $x \ge 0$, $y \ge 0$.

43 [Challenging] • Target: Using the horizontal line test • Cross-Lesson: 1.03 Rates of Change & Concavity

The graph of the polynomial function $f(x) = x^3 + ax^2 + bx - 6$ has a local maximum at $x = -3$ and a local minimum at $x = 1$. Find the values of a and b.

a. $a = -3, b = 9$

b. $a = 3, b = -9$

c. $a = -6, b = -3$

d. $a = 6, b = 3$

44 [Challenging] • Target: Composite functions with restricted domains • Cross-Lesson: 2.01 Exponential Functions & Growth Models

The graph of the polynomial function $f(x) = x^3 - ax^2 + bx - 6$ has a local maximum at $x = -1$ and a local minimum at $x = 3$. Determine the value of $a + b$.

a. 5

b. -5

c. 1

d. -1

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45 [Challenging] • Target: Composing functions • Cross-Lesson: 2.02 Logarithmic Functions & Properties

The function f is defined for all real numbers $xeq -2$ by $f(x) = \frac{4x-1}{x+2}$ What is the value of $f^{-1}(3)$?

a. -5

b. 3

c. 5

d. 7

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